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Laboratory Report (1,2,final):	Final

Please write your reports using this sectioned template, remembering your name and which laboratory session the report is from. There is no need to provide any of the formal structure of a full technical report, however relevant reference material should be included at the end of the section where it is referred to and it will be plagiarism checked. There is no limit on words or page length but you should also pay attention to the presentation of information and be selective in terms of figures and tables. Please pay attention to the marking scheme. And your summary feedback will be included

Marking Scheme (criteria against Learning Outcomes)	
Method & Results	Development of methods, models, collection of data and description of results.
Discussion & Analysis	Discussion of results, analysis of the data in terms of fundamentals. Conclusions.
Understanding & Design	Discussion of the best design with justification. Understanding and addressing the questions asked in the labs.
Presentation & Writing Style	Presentation of the information, style and choice of figures, clarity of written text

Marking criteria (weighting: session 1,2,final)	Feedback	Mark
Method & Results (1: 40%, 2: 30%, final: 25%)		
Discussion & Analysis (1: 40%, 2: 30%, final: 25%)		
Understanding & Design (1: 20%, 2: 30%, final: 25%)		
Presentation & Writing Style (1: 0%, 2: 10%, final: 25%)		
Total overall		

Section 1: Methods

The analysis was performed using COMSOL by employing the Finite Element Method (FEM) to solve the coupled Navier-Stokes equations for laminar flow and transport of diluted species (convection-diffusion). The problem space was modelled as a 3D microfluidic chamber with defined inlet and outlet ports. Boundary conditions were established as follows:

- **Inlet:** Defined by a constant flow rate (fr) and analyte concentration (C_0). Position “wp1” at $x = 0$.
- **Outlet:** Pressure boundary condition ($p = 0$). Position “wp3” at $x = 1$.
- **Walls:** No-slip boundary condition ($u = 0$) to enforce laminar flow characteristics.
- **Sensor Surface:** Modelled using a surface reaction rate $r_{ads} = k_{ads} * C_x$ where k_{ads} is the forward rate constant, and C_x is the concentration at the boundary.

Parameters

Name	Expression	Value	Description
D	$2e-9[m^2/s]$	$2E-9 \text{ m}^2/s$	Diffusion constant for Sol...
fr	$0.1666[ul/s]$	$1.666E-10 \text{ m}^3/s$	Inlet flow rate
alpha	$0.5[(m^3/mol)^{-2}]$	$0.5 \text{ m}^3/mol^2$	Viscosity c^2 -term prefa...
k_ads	$1e-6[m/s]$	$1E-6 \text{ m/s}$	Forward rate constant
inlet_rad	$0.1[mm]$	$1E-4 \text{ m}$	Inlet/Outlet Radius
inlet_length	$1[mm]$	0.001 m	Inlet Length
c0	$100[mol/m^3]$	100 mol/m^3	Inlet concentration (0.1 M)

Figure 1 Initial global parameters

In this lab, a two-fold simulation is done. The equations given define how the analyte moves through the fluid and how it is consumed by the sensor.

First is the steady state (Creeping flow) simulation, which is defined by Fick's first law, which relates the diffusive flux to the gradient of concentration [4].

It is denoted by $J_D = -D\nabla C_x$, where

- J_D is the diffusion flux, which measures how much analyte flows through a unit area during a unit time interval.
- D is the diffusion coefficient
- ∇C_x is the concentration gradient, which changes with characteristic length.

The first simulation is independent of time.

After that, the flow is introduced into the system, which is denoted by $J_u = uC_x$ [4]. It is better understood by Fick's second law, which adds a directional term that accounts for the fluid's motion driven by the pump. Therefore, the total flux is denoted by $N = J_t = -D\nabla C_x + uC_x$.

The sensor detects the flux on that surface as $-n * N = -k_{ads} * C_x$ which is equivalent to

$$-n * (-D\nabla C_x + uC_x) = k_{ads} * C_x$$

In a flow-based system, the uC_x term brings fresh analyte to the sensor surface continuously. This replenishment reduces the thickness of the depletion layer and allows the sensor to reach a stable, steady state signal rather than decaying as it would in a static chamber.

Mesh Construction

To ensure numerical stability, a "finer" mesh was generated. This was particularly critical at the input/output ports and the sensor surface, where steep concentration gradients and velocity changes occur.

Experimental Design

The study investigated the impact of five primary variables:

1. **Initial Concentration (C_x):** Ranging from 1 to 100 mol/m³ to assess sensor linearity.
2. **Rate Constant (k_{ads}):** Varied between 10⁻⁸ and 10⁻⁴ m/s to observe the transition between reaction-limited and diffusion-limited regimes.
3. **Diffusion constant (D):** In the range of 2E-9 to 2E-15
4. **Chamber Geometry:** Scaling the height (h) by introducing elliptical voids to optimize mass transport to the sensor surface in the range of c-semiaxis = 0.5 to 1.
5. **Flow rate (fr):** Varying it in the range of 0.1 μ l/s to 0.5 μ l/s.

Based on the context of COMSOL simulations, this expression represents a global post-processing operation used to calculate the integral of a flux component over a boundary or domain:

$$\text{Global Evaluation 1 (intop1}(-nz * \text{tds.ntflux}_c))$$

- **Intop1(...):** This is a global integration operator. This tells COMSOL to calculate the integral of the expression within the (...) over the specific selected geometric entity.
- **(- negative sign):** This is a mathematical scalar multiplier, which is used to ensure the sign of the result matches the desired physical convention.
- **nz:** This represents the z-component of the surface normal vector. In transport physics, when calculating the flux across a surface, the dot product of the flux vector and the surface normal vector is needed.
- **tds.ntflux_c:** This represents the total mass of the species "c" normal to the boundary in the tds interface.

Section 2: Results

1. Given both convection and diffusion, consider how your sensor geometry affects sensor response. Present your optimal sensor design.

Optimal Design:

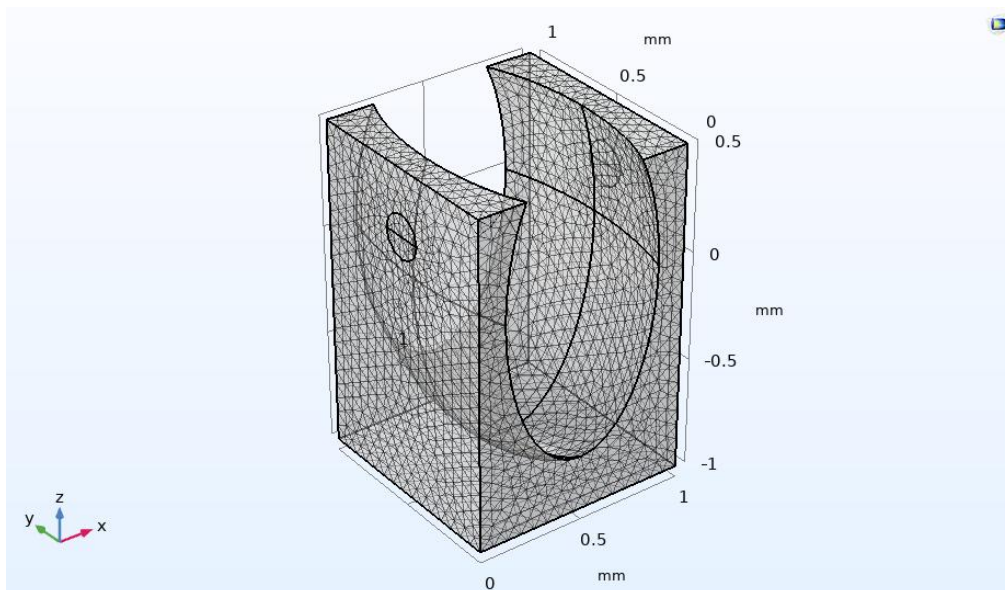


Figure 2 Optimized geometry for better sensitivity

Key modifications:

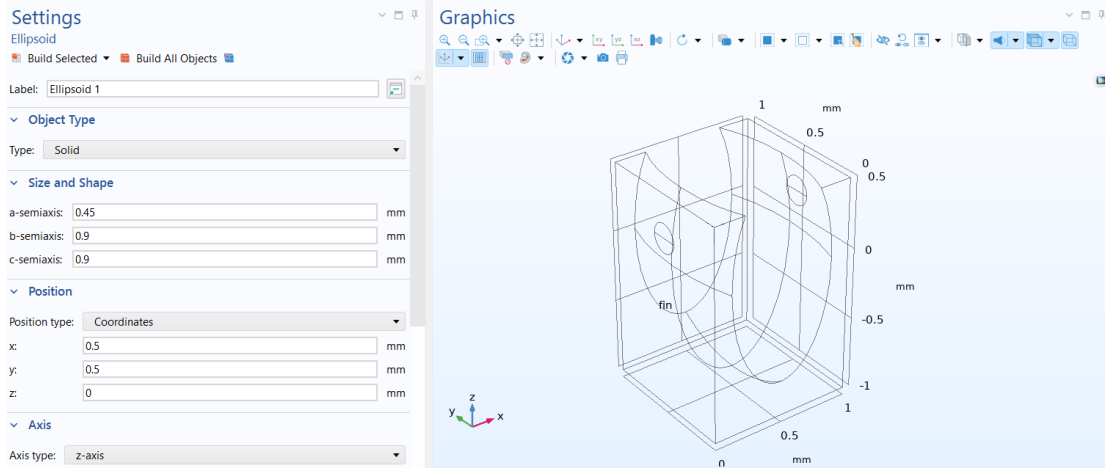


Figure 3 Position of the ellipsoid to change the characteristic height

- An ellipsoid is used to make a void in the center of the design, which makes a parabolic path so that the fluid may flow close to the boundaries and the sensor.
- When the ellipsoid is cut from the structure, the aspect ratio of the channel changed [5]. A standard mesh might become stretched or skewed in a thin channel. Applying a finer mesh ensures that the elements remain high-quality, preventing geometric errors. For low diffusion coefficients, the concentration changes happen over extremely small distances. A finer mesh is mandatory here to prevent unwanted oscillations in the results.

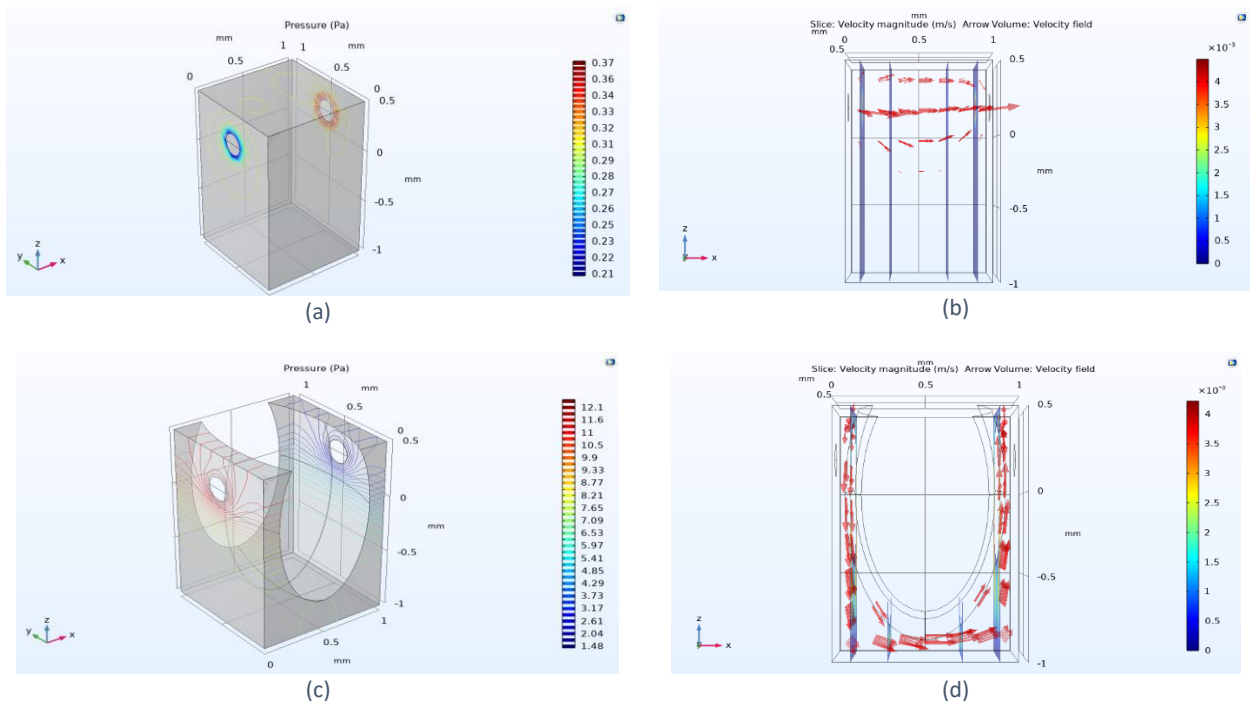


Figure 4 (a) Baseline static pressure contour plot. The linear pressure drop from inlet to outlet reflects the hydrodynamic resistance of a uniform cross-section chamber before optimization. (b) Initial 3D velocity magnitude distribution within the rectangular microfluidic chamber (c) Optimized pressure distribution showing a significant increase in gradient. (d) Velocity magnitude profile following geometric optimization.

In the initial design, the fluid followed a path of least resistance through a large, open rectangular volume [1][3]. By subtracting an ellipsoid from the upper surface, the fluid was effectively squeezed.

- **Impact:** This constriction forces the bulk of the fluid streamlines to pass closer to the sensor surface which reduces the distance the analyte must travel via diffusion to be detected, effectively minimizing the "dead zones" where analyte might linger.

The results show that the pressure gradient nearly doubled.

- **Impact:** This increase in pressure at the inlet is caused by the reduced cross-sectional area. According to the analogy of an electrical circuit ($V=IR$), decreasing the chamber height or volume while keeping the flow rate constant increases the hydrodynamic resistance (R_H). While this requires more work from the pump, it results in a "flatter" and more efficient flow of fluid from inlet to outlet, directly improving sensor sensitivity [1][3].

By bringing the fluid closer to the surface, P_e has been increased near the boundary. This helps transition the sensor into a more stable diffusion-limited regime at high k_{ads} , where the signal is dictated by the predictable arrival of analyte rather than erratic surface kinetics [1].

- The higher velocity in a thin channel may potentially damage the sensor surface.

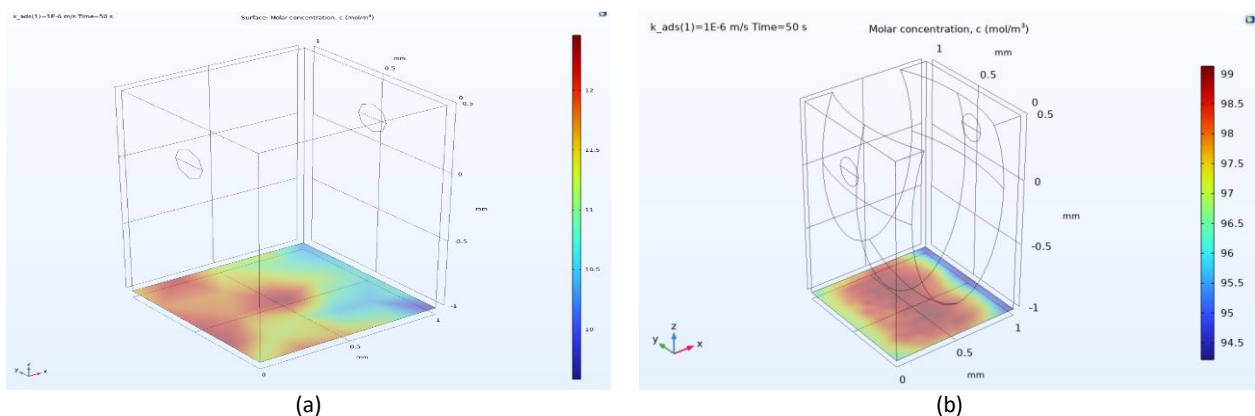


Figure 5 (a) Expansion of the depletion zone into the bulk fluid; the signal begins to decay as the local concentration at the sensor decreases. (b) Attainment of a steady-state concentration profile; the forced convection effectively balances the surface consumption rate.

From Figure 5(b), it is observed that the concentration stops changing significantly after 50 seconds in the optimized geometry, which happens because it has achieved dynamic equilibrium between two competing physical processes:

- **Analyte Consumption (Sink):** The sensor surface is continuously removing analyte at a rate determined by k_{ads} .
- **Analyte Replenishment (Source):** The optimized, high-velocity flow path continuously brings "fresh" bulk concentration from the inlet to the sensor.

In the initial geometry, the flow path was too far from the sensor, allowing the depletion zone to keep growing larger. In the Optimized Geometry, the constriction ensures that the rate at which the pump delivers new molecules to the surface exactly matches the rate at which the sensor consumes them. This creates a steady-state condition where the concentration profile remains fixed in space.

The optimized geometry increases localized advection, allowing the system to reach a steady state within approximately 50 seconds. At this point, the rate of analyte replenishment from the pump matches the rate of consumption by the sensor, providing a stable, long-term signal that is essential for accurate real-time monitoring.

2. Does the sensor output signal provide a means of determining the concentration in the incoming sample?

The primary goal of this sensor is to correlate an electrical signal back to a bulk concentration (C_o). Based on the analysis of Table 1 data: It shows a linear relationship between the inlet concentration (C_o) and the sensor flux while keeping the diffusion coefficient (D) constant at $2 \times 10^{-9} \text{ m}^2/\text{s}$. The most crucial observation is that for a constant k_{ads} , the output flux scales perfectly with C_o .

• For $k_{\text{ads}} = 1 \times 10^{-4} \text{ m/s}$, the steady-state flux values at $t=300\text{s}$ are:

- $C_o = 100 \text{ mol/m}^3$, the flux is $\approx 2.42 \times 10^{-10} \text{ mol/s}$
- $C_o = 10 \text{ mol/m}^3$, the flux is $\approx 2.42 \times 10^{-11} \text{ mol/s}$
- $C_o = 1 \text{ mol/m}^3$, the flux is $\approx 2.42 \times 10^{-12} \text{ mol/s}$

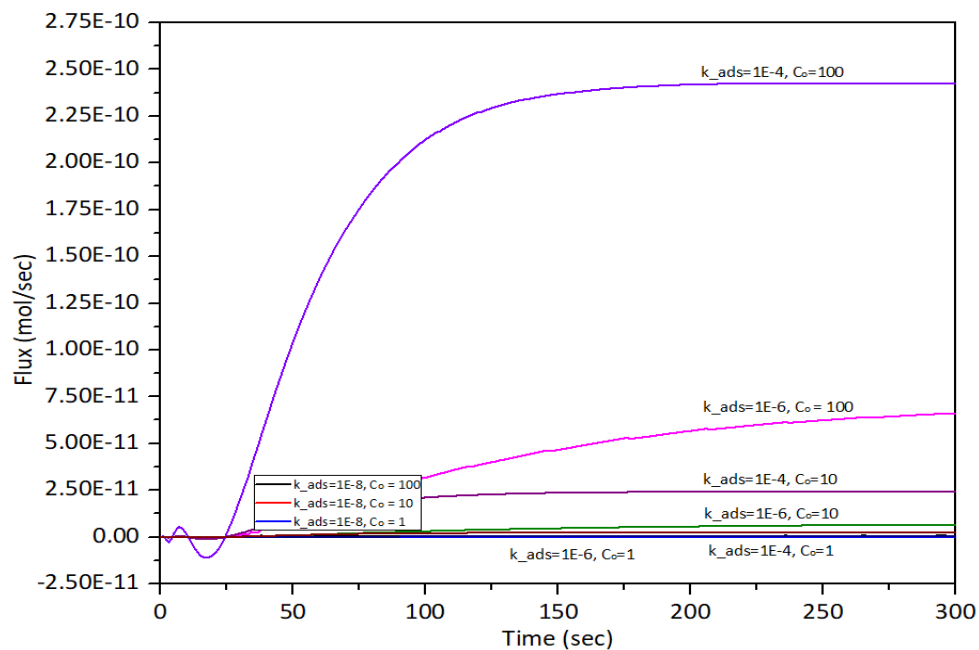


Figure 6 Effect of varying C_o and k_{ads} on sensor response

Conclusion: Yes, the sensor output signal provides a reliable means of determining unknown concentration. It also implies that the sensor does not suffer from saturation effects within this concentration range. Since the signal scales linearly with C_o , the concentration in an incoming sample can be determined by using a simple sensitivity factor $S = \frac{\text{Signal}}{C_o}$.

Therefore, if the signal is $2.42 \times 10^{-11} \text{ mol/s}$ for $C_o = 10 \text{ mol/m}^3$, it can be said that the sample concentration is 10 mol/m^3 .

3. From the perspective of the sensor output signal, what is the effect of the rate constant for the surface reaction, and how would this affect your ability to determine concentration in a real sensor?

The variable of diffusion constant " D " to illustrate how the rate constant interacts with physical transport is introduced.

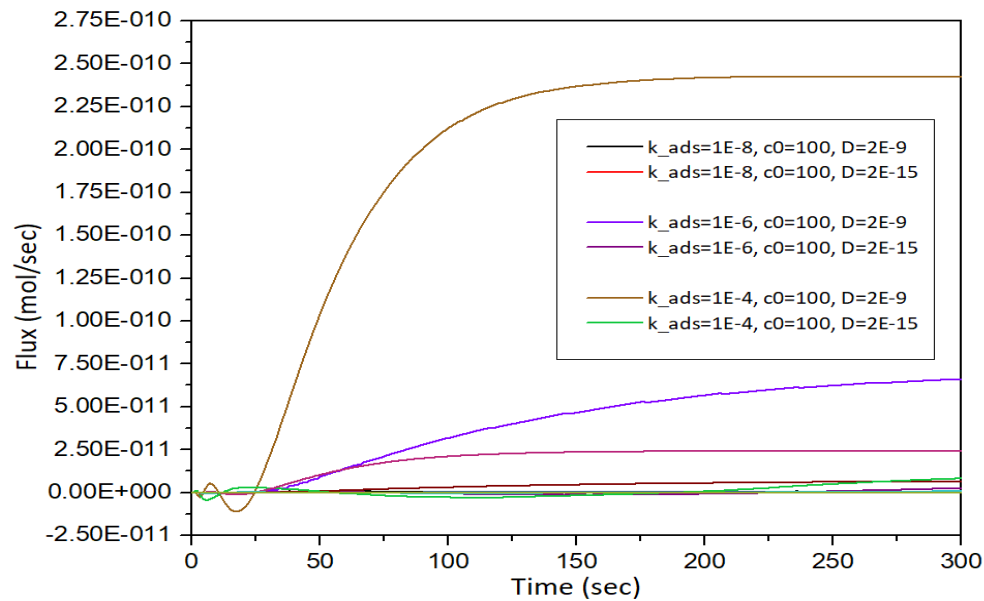


Figure 7 Effect of varying k_{ads} , C_o , and D on sensor response

The simulation results in Figure 6 and 7 highlight two distinct regimes of operation:

- **Low k_{ads} (e.g., 10^{-8} m/s):** The system is reaction-limited. The flux is very low ($\sim 5.7 \cdot 10^{-13}$ mol/s for $C_o = 100$). Here, the molecules are reaching the sensor surface via diffusion, but the surface is slow to react with them. Because the analyte is not being consumed fast enough to create a depletion zone, it becomes unreliable for concentration measurement [2].
- **High k_{ads} (e.g., 10^{-4} m/s):** The system becomes diffusion-limited. The signal is much higher ($\sim 2.42 \cdot 10^{-10}$ mol/s) because the surface is so reactive that it consumes the analyte molecule as soon as they arrive. This creates a depletion zone near the bottom of the channel where the surface concentration becomes zero. The sensor's performance is no longer limited by chemistry, but by how fast the flow can deliver new molecules to the surface [2].

When D is extremely low, the flux drops dramatically even if the k_{ads} is high. Molecules with low D are trapped in the streamlines of the fluid. Even though the flow is bringing them into the chamber, they cannot jump across the streamlines to reach the sensor surface. It also shows that if " D " drops from $2 \cdot 10^{-9}$ to $2 \cdot 10^{-15}$ m²/s, the signal crashes from 10^{-10} to 10^{-12} mol/s.

Real-world Impact:

In a practical application, k_{ads} is often unknown or a variable parameter. It can be affected by temperature, pH, etc., and it becomes difficult to distinguish whether a low signal is due to a low sample concentration or a "dirty" sensor with a lower effective k_{ads} .

- **For a reaction-limited sensor:** The signal is heavily dependent on the quality of the sensor surface. A 10% drop in surface activity k_{ads} results in a 10% drop in signal, leading to an incorrect concentration reading.
- **For a diffusion-limited sensor:** The signal depends primarily on (D) and the flow velocity (u). Since D and u are more stable and are less sensitive to small changes in k_{ads} , operating in the diffusion-limited regime makes the sensor more robust and easier to calibrate for real-world samples.

4. Is the sensor under flow conditions better than a sensor system with only diffusion?

A comparison between the results of Lab 2, where there was static diffusion and Lab 3, where there was convection, has to be made.

Comparison:

- **Diffusion Only (Lab 2):** In this case, the sensor signal starts high and eventually decays towards zero. It occurs because the finite volume of the analyte in the chamber is consumed, and as the distance to fresh analyte increases, the diffusion flux drops. This is only useful for point-of-care disposable tests where a single peak value is measured.
- **Flow Conditions (Lab 3):** In this system, the analyte is constantly replenished at the inlet. As seen in Figure 6, the signal reaches a stable steady state by $t = 100$ s. Unlike the static sensor, the flow-based sensor can monitor concentration changes in real-time. If the inlet concentration were to change at $t = 400$ s, the sensor would respond and reach a new steady state. This also suggests that the P_e is significantly greater than 1.

Conclusion: The flow system is superior for continuous monitoring of most lab-on-a-chip applications. It provides a stable, repeatable signal by eliminating the depletion problem found in Lab 2. By maintaining a constant supply of the analyte, it is ensured that the sensor operates at its maximum potential flux, providing a continuous and far more powerful diagnostic signal. It prevents the signal from dropping off over time, allowing for long-term measurements of the sample concentration.

Section 3: Discussion

In microfluidic systems, transitioning from pure diffusion (Lab 2) to involving both convection and diffusion, the physical geometry of the sensor chamber becomes a critical detriment to its performance. The sensor response is heavily influenced by the chamber height and the location of the sensor.

- In a pure diffusion system, the analyte is eventually exhausted, but in a flow-based system, a dynamic equilibrium between the bulk transport, i.e. convection and surface reaction is established [3]. It was previously observed that reducing the chamber height increases R_H ($R \propto \frac{1}{h^3}$), requiring higher pressure to maintain the same flow rate, which leads to a faster decay in signal (Lab 2).
- The "diffusion time" for an analyte molecule to reach the sensor from the centre of the flow is proportional to $\frac{h^2}{D}$ [1]. Taking this into account, in Lab 3 the flow brings fresh analyte to the sensor, but a depletion layer forms near the surface, that's why, a shallower chamber is significantly more efficient because by reducing the height, it is ensured that a higher percentage of the analyte molecules passing through the chamber actually come into contact with the sensor surface rather than being swept away by the velocity field. This increases the flux of the system.

Since this system is the combined form of Lab 1 and Lab 2, the characteristic length and fluid velocity is also calculated to be the same. Therefore,

$$R_e = \frac{\rho UL}{\mu}$$

$$\rho = 1000 \frac{kg}{m^3} \quad \mu = 1 \text{ mPa} \cdot s \quad U = 0.01142 \text{ m/s}$$

Taking $L = 100 \mu m$

$$R_e = \frac{1000 \frac{kg}{m^3} * 0.01142 \frac{m}{s} * 1 * 10^{-4} m}{1 * 10^{-3} Pa \cdot s} = 1.142$$

Since the calculated value is much closer to 1, viscosity still dominates over inertial forces, and the system will have a laminar flow.

Taking these values and $D = 2E-9$, the P_e comes out to be much greater than 1, proving the point of a superior real-time monitoring based on the flow-based system, which provides a stable, steady-state signal.

Table 1 Pressure at the inlet; variation with respect to flow rate

Flow rate ($\mu\text{l/s}$)	Pressure (Pa)
0.1666	7.59
0.2666	12.1
0.3666	16.7
0.4666	21.2
0.5666	25.8

It is observed that for a flow rate of $0.1666 \mu\text{l/s}$, the signal is very weak and decays over time. When the flow rate is increased to $0.3666 \mu\text{l/s}$, the signal reaches a steady-state within 50s with a strong and linear response to concentration, and for flow rate $> 0.4 \mu\text{l/s}$, the pressure gradient significantly increases, which risks mechanical failure.

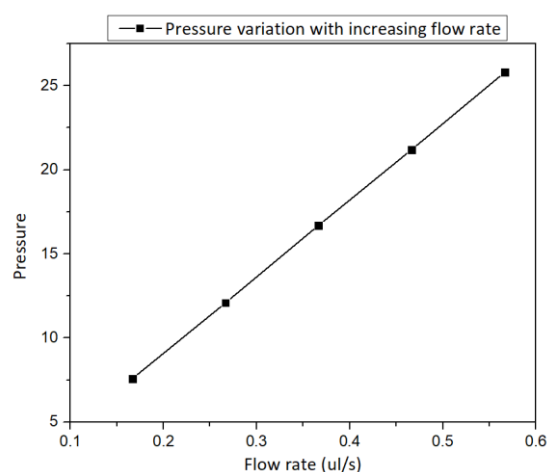


Figure 8 Graphical depiction of Table 2

By calculating the slope of Figure 4, the value of R_H can be calculated. The strictly linear trend indicates that the system is operating in a Linear Viscous Regime, and the R_e of 1.142 (Lab 1) makes it a highly predictable design.

Section 4: Summary of Conclusions

- On the basis of the sensor characterization it is evident that the system consistently operates within a low R_e regime, guaranteeing a stable laminar flow profile. This predictable behaviour is essential to make the sensor accurate because it prevents turbulence.
- Geometric optimization was critical; the use of an ellipsoidal subtraction and chamfering of sharp 90° corners effectively removed stagnant “dead zones” where the analyte was previously trapped. This reduction in the chamber height causes fluid streamlines to get closer to the sensor, causing the concentration boundary layer to squeeze and the amount to the sensor to rise. Although this optimization almost doubled the hydrodynamic pressure gradient and resistance, this is an unavoidable sacrifice for such a substantial increase in sensitivity and speed.

- In addition, the study revealed that an operation under a diffusion-limited regime is optimal for quantitative diagnostics, whereas a regime with high surface kinetics results in a weak dependence of the signal on bulk concentration. Most importantly, the system came to a dynamic equilibrium thanks to the shift from static to flow-based transport.
- The optimized flow geometry eliminates the continuous degradation of the signal because of the depletion of the reagents in the static system, and provides a stable steady-state within about 50 seconds that allows for a consistent signal for real-time monitoring. The results presented here give further evidence that for a reliable and highly sensitive microfluidic diagnostic device, a streamlined, high-velocity flow path is essential.
- Based on the calculated R_e , P_e , and the relation between the flow rate and the pressure gradient (Figure 8), this system gives us a predictable design flow.

Section 5: Notes on answers to questions and your thoughts

References

- [1] H. Bruus, *Theoretical microfluidics*. Oxford: Oxford University Press, 2008.
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- [5] S. Franssila, *Introduction to microfabrication*, 2nd ed. Chichester, West Sussex, England: John Wiley & Sons, 2010.